

JAIST — Taniike Lab

Integrating OCM Literature Data into Lab-Scale ML Models

A Walkthrough of Five Transfer-Learning Approaches

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The Problem — Two Datasets, Three Systematic Differences

Chapter 1

Our Lab Data

Samples	89,074
Year	2025
Preparation	Impregnation only
Mean Y(C2)	5.25%
Conditions	Fixed, one lab

Published Literature Data (≤ 2019)

Samples	3,852
Year	1982 - 2019
Preparation	15+ methods
Mean Y(C2)	8.67%
Conditions	40 years of diverse labs

Can we use literature to improve our model — without hurting it?

1. Label shift

Literature mean is 3.42 pp higher ($t = -32.2$, $p < 10^{-200}$) — a systematic offset from publication bias and optimised conditions.

2. Covariate shift

78.5% of literature samples describe chemistry our lab has never tested.

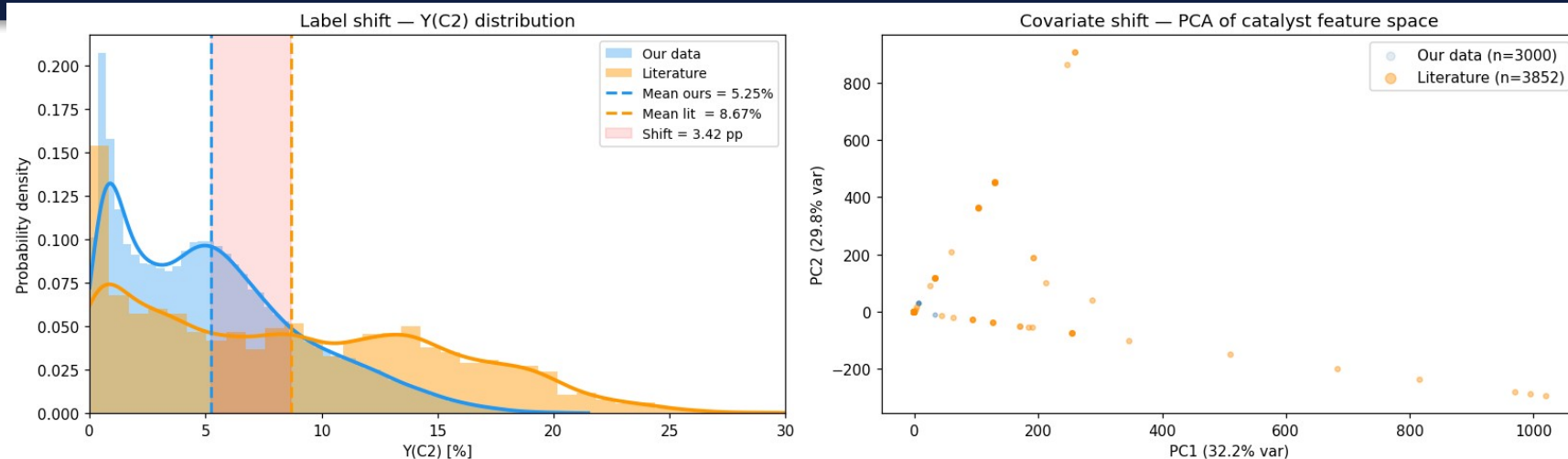
3. Publication bias

Literature is skewed, not just shifted — failed experiments are rarely reported, over-weighting high yields.

Why ML? The catalyst search space (65 elements $\times$ varying loadings $\times$ temperature) is too large to screen physically. A surrogate model guides experiments toward high-yield compositions.

Setup — Label Shift & Covariate Shift

Chapter 2



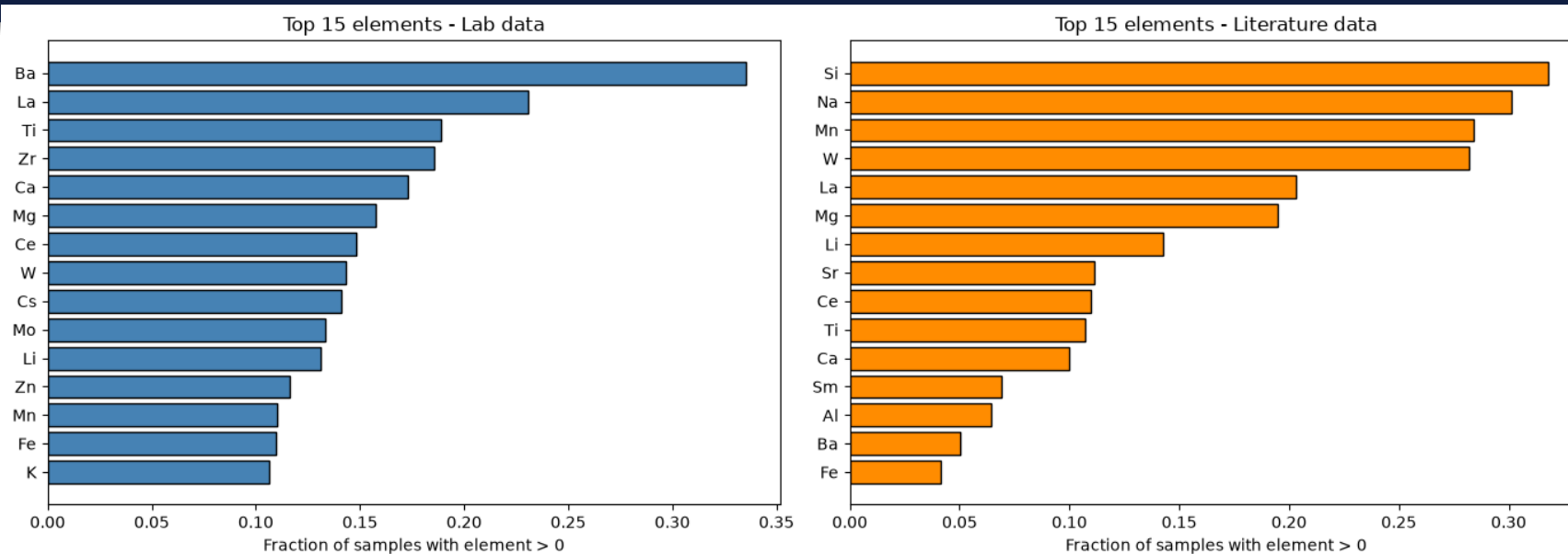
Two panels, two shifts

- Left: yield KDE — blue peaks 3–4%
- Orange peaks 8–9% (3.42 pp gap)
- Red band = systematic gap ($t = -32.2$)
- Orange right tail = publication-bias skew
- Right: PCA of 67 features → 2D
- Orange outside blue = 78.5% OOD

Both signals — yield level AND chemistry composition — warn against naive concatenation (confirmed slide 6).

Setup — Which Elements Differ?

Chapter 2



Reading the chart

- Top 15 elements by usage frequency
- Different dominant elements each side
- Different active phases & promoters
- Our lab: focused 2025 programme
- Literature: 40 yrs, many groups

This is what '78.5% out-of-distribution' looks like in plain terms — literally different elements and preparation methods.

Selective filtering methods keep only literature whose element profile resembles our experimental programme.

How We Measure Success — Catalyst-Grouped 5-Fold CV

Chapter 3

The validation rule

- The 89,074 rows are only 917 DISTINCT CATALYSTS
- Folds split by catalyst — all rows of a catalyst in one fold
- Validation fold = catalysts the model has never seen
- Literature never appears in validation

Why grouped, and not by row

- A row split puts the SAME catalyst in train and test
- The model then recalls catalysts instead of predicting new ones
- Prof. Taniike raised exactly this — grouped CV is now the default

Primary metric: catalyst-level

Spearman on each catalyst's best yield, and enrichment@10%.
Row RMSE is secondary — 19.9% of its variance is unreachable.

5-fold split by CATALYST (917 groups)



+ literature (training only, never validation)

Rotate the held-out fold 5 times. Every number on the following slides uses this protocol; row-level figures appear only where they are labelled as historical.

Baseline + Why Naive Merging Fails

Chapter 4

Step 1 — establish baseline (ROW-LEVEL CV, historical)

Method	CV RMSE
Baseline (lab data only)	2.133

Row-level protocol — kept only for comparison with the published numbers. Under catalyst-grouped CV the baseline is 2.9425.

Step 2 — try the obvious

Method	CV RMSE
Baseline	2.133
Naive merge (3,852 lit rows)	2.241

+5.1% worse than baseline (row-level). Re-tested at catalyst level, a direct merge again sits below the composition-only control (0.7577 vs 0.7606).

Why did it fail?

Model sees two catalysts with nearly identical composition:

- One labelled 5.2% (our lab)
- One labelled 8.7% (literature)

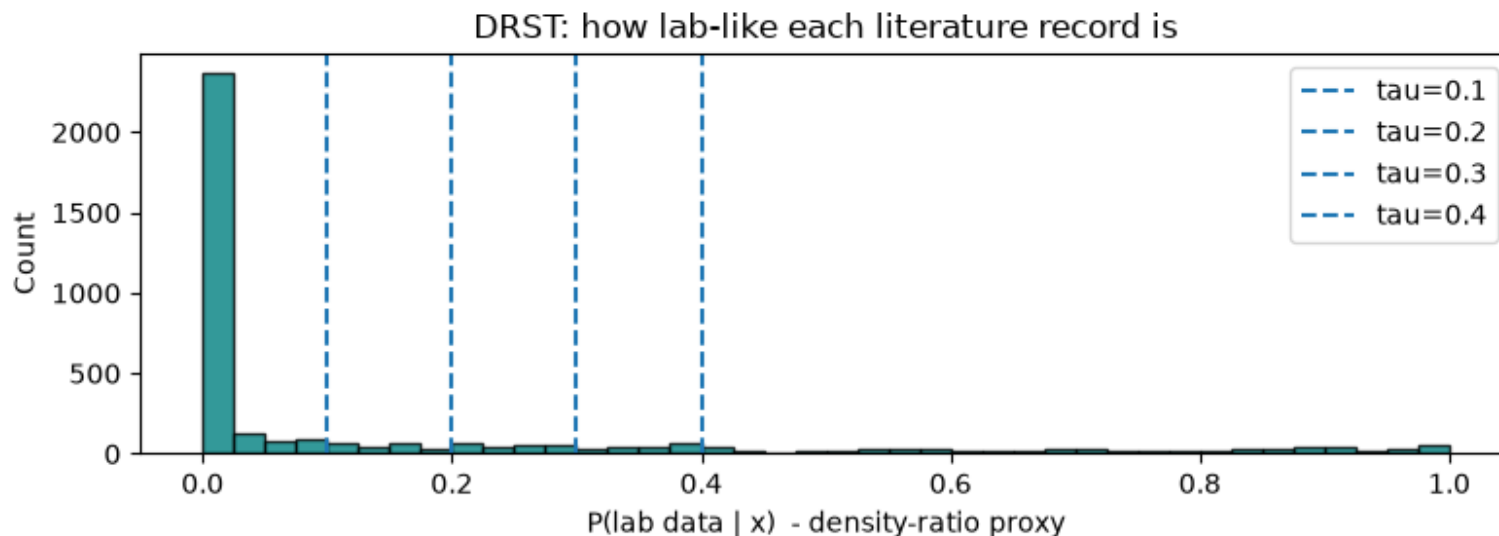
No feature explains the gap — flow rate, gas ratio, etc. are absent. The model hedges and predicts both badly.

The 3.42 pp offset is a SYSTEMATIC signal the model cannot account for — and incorrectly tries to fit. Mixing the two datasets can perform WORSE than using ours alone.

⇒ We need to be much more selective.

DRST — Filtering by Chemical Similarity

Chapter 5



Histogram of $P(\text{ours} | x)$ for every literature sample. Dashed lines = candidate thresholds.

How it works

- Train logistic classifier: ours=1, lit=0
- Score each lit sample: $P(\text{ours} | x)$
- Keep samples scoring $\geq \tau$

$P(\text{ours} | x)$ is a similarity score: high = looks like our chemistry, low = foreign. We keep only literature scoring above τ .

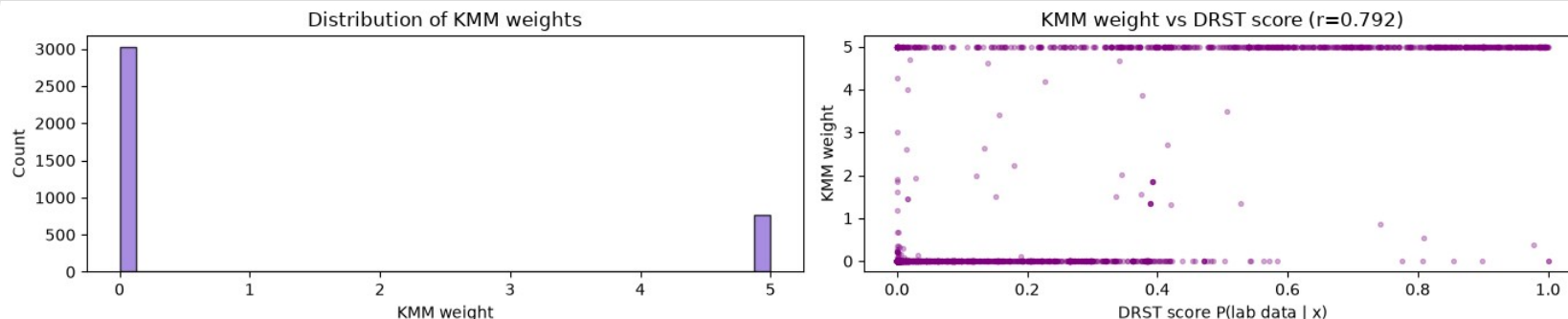
Threshold sweep

τ	kept	RMSE
$\tau = 0.10$	1168 kept	2.068
$\tau = 0.20$	987 kept	2.034
best τ	~200 kept	2.127
$\tau = 0.40$	543 kept	2.045

Unsolved: hard cutoff is arbitrary — a sample at 0.29 is fully discarded; 0.31 gets full weight.

KMM — Soft Weights Instead of a Hard Filter

Chapter 6



How it works

- Each lit sample gets weight $w_i \in [0, 10]$
- RBF kernel: $K(x, x') = \exp(-\|x - x'\|^2 / 2\sigma^2)$
- Solve QP: $\frac{1}{2} w^T K_{ss} w - \kappa^T w$
- σ via median heuristic (parameter-free)

KMM picks weights so the weighted literature cloud matches lab data cloud. Samples inside our chemistry earn weight; those outside fade to zero.

Result

KMM CV RMSE	2.261
Δ vs baseline	+6.0% (worse)
Near-zero weights	78.5%
Agreement w/ DRST	$r = 0.79$

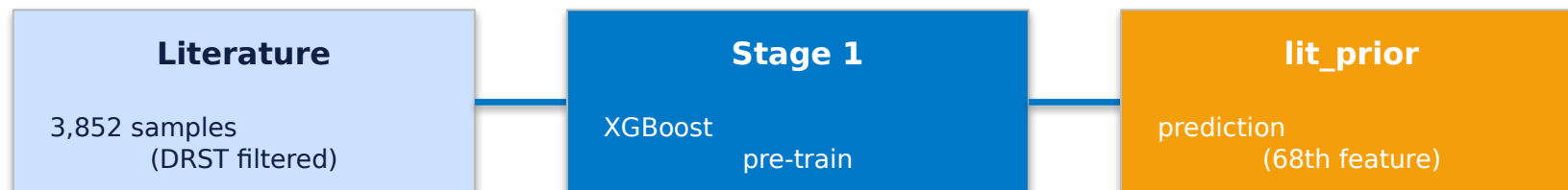
Reassuring: two unrelated methods flag the same 78.5% of samples $\Rightarrow$ real signal, not artifact.

Still unsolved: literature labels remain in training, so the 3.42 pp offset still bleeds in.

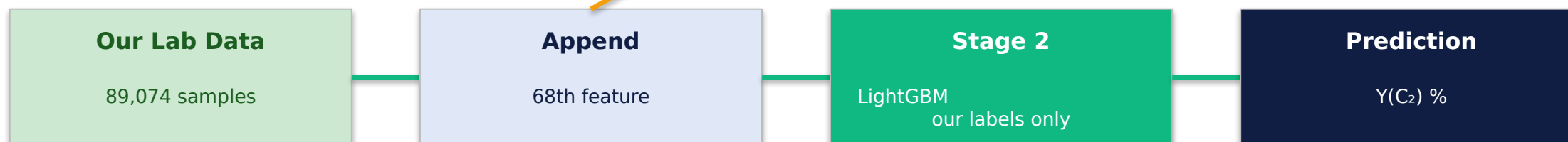
Prior Feature Transfer — The Method That Exposed the Leak

Chapter 7

STAGE 1 — Pre-train on Literature Chemistry



STAGE 2 — Fine-tune on Our Labels Only

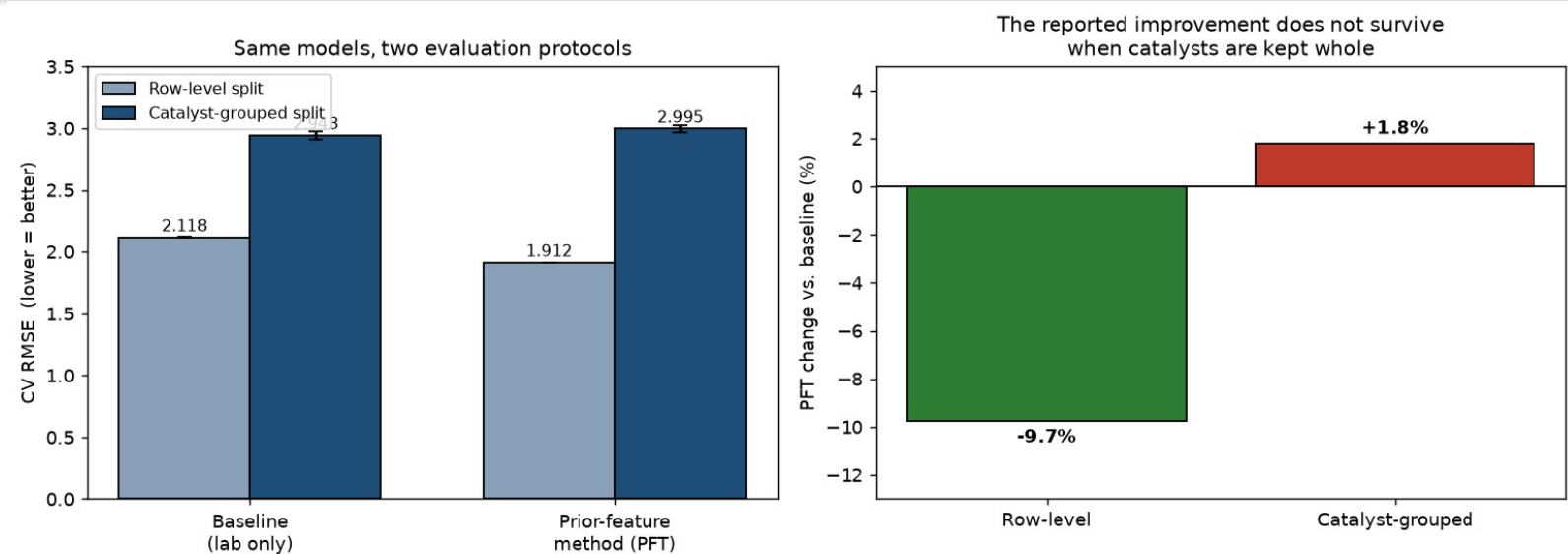


The design was sound. The evaluation was not.

Literature enters as a FEATURE, never as a label, so the 3.42 pp offset cannot corrupt the training loss. That reasoning still holds. But Stage 1 was trained on literature TOGETHER WITH the lab training rows, and under a row-level split those rows included the test catalysts — so the prior feature partly carried each test catalyst's own measured yields. The gain was recall, not prediction. Ruling this out is what produced the protocol on slide 5; the mechanism is named in `ocm_eval.stage1_data()`.

Results — The Same Models Under Two Protocols

Chapter 8



Row-level CV — HISTORICAL, leaked

Method	RMSE	Δ
Baseline	2.1184	—
PFT (filtered)	1.9120	−9.7%
PFT (all lit.)	1.9194	−9.4%

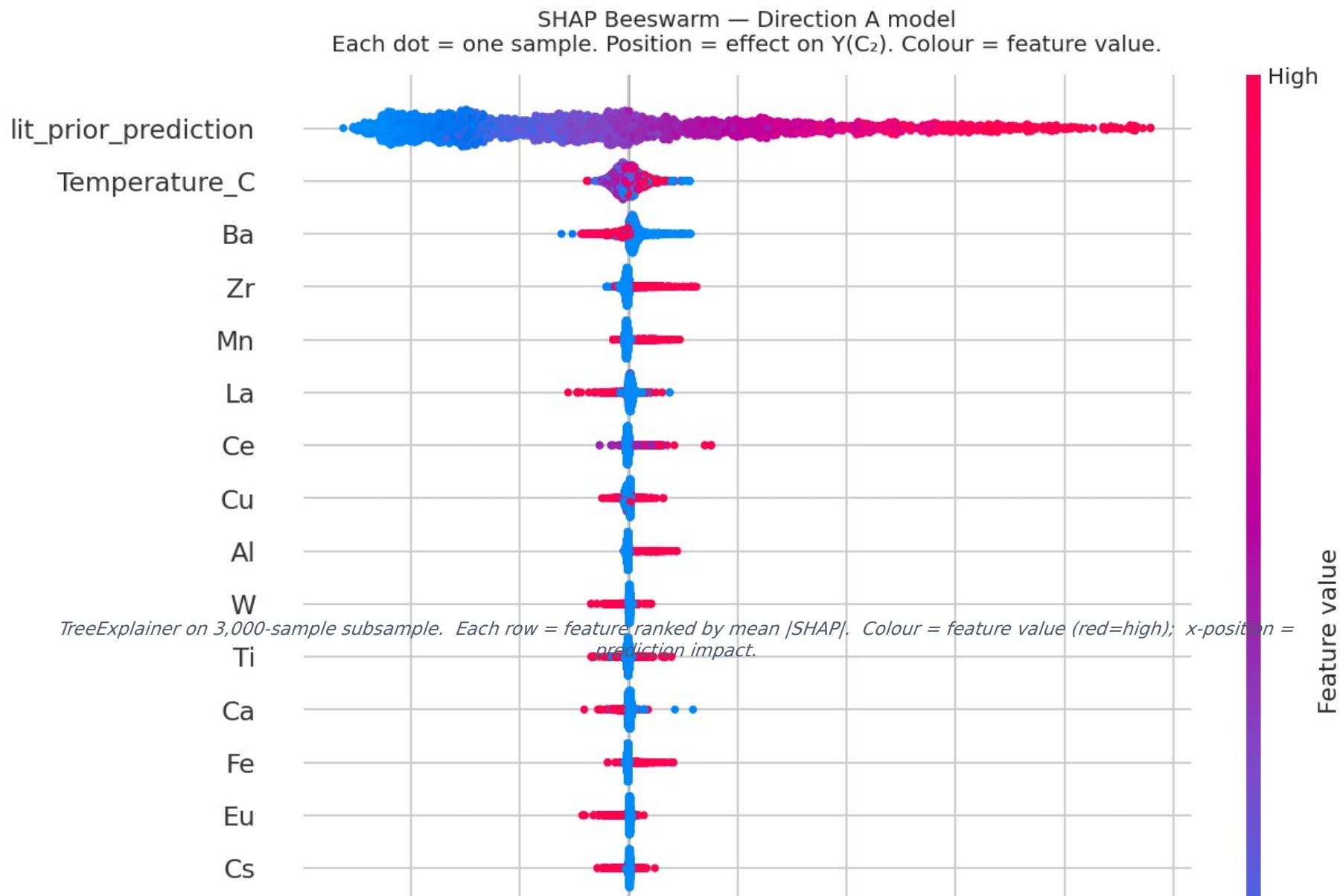
Catalyst-grouped CV — DEFAULT

Method	RMSE	Δ
Baseline	2.9425	—
PFT (filtered)	2.9955	+1.8%
PFT (all lit.)	2.9817	+1.3%

Same models, same data, two protocols. The published −9.7% becomes +1.8% WORSE once a catalyst cannot appear in both train and test. The difference is leakage, not modelling.

SHAP — Feature Importance Beeswarm

Chapter 8



Key signals

- #1: lit_prior — transfer used
- Temp $\uparrow \rightarrow$ yield $\uparrow$ (correct OCM)
- Ba, Mn, La, Ce: net positive
- Li, K: mixed / suppressive

The model uses real OCM chemistry — transfer learning is not dead weight.

*See next slide for residual breakdown
→*

SHAP — Ceiling Effect & Model Limits

Chapter 8

Residual breakdown by yield range

Y(C ₂) range	RMSE
0 – 6%	1.43 – 1.48
6 – 10%	1.95
10 – 15%	2.58
> 15%	4.70 (–4.2 pp bias)

Root cause: GHSV, CH₄/O₂ ratio, pressure absent from the feature set — a data limit, not a model failure.

What SHAP confirmed

- #1: lit_prior_prediction — transfer learning actively used
- Temperature: high T → yield up (correct OCM physics)
- Ba, Mn, La, Ce — known active phases and promoters
- Li, K — mixed / suppressive (can over-reduce surface)

The ceiling is NOT a modelling failure — it is a DATA failure.
Missing columns (GHSV, CH₄:O₂, pressure) explain why Y>15% catalysts exist.
No algorithm can compensate for absent input features.

Fix the data → the model automatically improves. That is the single highest-impact next step.

Statistical Rigour & Honesty

Chapter 8

The headline was coverage-inflated

- Cells run further contain better yields, so a score over all
- 917 catalysts is partly a record of which runs were finished.
- All 917: Spearman 0.767 enrichment 4.35×
- Equal effort (771): Spearman 0.724 enrichment 3.77×
- Coupling removed BY MEASUREMENT: $\rho(n_rows, max) +0.293 \rightarrow +0.003$
- 300 random 771-subsets give 0.767 [0.756, 0.780] — 0.724 sits below that interval, so the drop is real, not a smaller-sample effect.

The sharpest test we could devise

- We retrained the model to predict HOW MANY measurements a catalyst received. It never sees a yield.
- That model reaches Spearman 0.400 against observed max yield...
- ...but its enrichment is 0.87× — no better than random.
- Rank correlation is partly purchasable from experimental effort.
- Enrichment is not. That is why enrichment is the primary metric.

And the open question is bounded: ranking catalysts by their observed FLOOR instead of their ceiling shares only 7 of the top 20, but a ceiling-trained model loses just 0.017 Spearman against floor ground truth — less than seed-averaging alone buys — and never drops below 4.02× enrichment. The answer changes the labels, not the decision.

Critical Analysis — Per-Method Review

Chapter 9

Honest critique of each method

Method	Main limitation
Naive merge	Negative control — makes the model worse
DRST	Hard cliff; discards ~80%; τ tuned on the reported CV
KMM	$O(n^2)$ kernel; σ -sensitive; labels still in loss; ties DRST
Prior FT	Best — Stage-1 on 782 rows → high variance; two models

Cross-cutting limits

- All methods evaluated on our-data CV only; OOD tested for Prior FT alone
- Point predictions — no uncertainty estimates to guide experiment selection
- $Y > 15\%$ ceiling is DATA-bound (missing GHSV, $\text{CH}_4:\text{O}_2$, pressure), not model-bound

Key insight: the ceiling effect and much of the label shift share the same root cause — missing operating-condition features. Fix the data; the models follow.

What's Next — Priority-Ordered

Chapter 9

1

Ask JAIST for the reaction-condition columns **Highest impact**

One email. Converts 19.9% of currently unreachable variance into modellable signal and turns 917 training examples back into 89,074.

2

Ask whether the 27 slots are conditions or a stream **Highest impact**

Provably unanswerable from the file: every cell is stored sorted by yield, so row order records rank, not acquisition sequence. Only the lab can settle it.

3

Ask why grid coverage is incomplete

186 catalyst-temperature cells absent; only 811 of 917 catalysts have all five temperatures. Decides whether the bias is correctable or itself informative.

4

Re-scope the campaign before reactor time is spent

20 runs per catalyst instead of 135 reproduces the full ranking at $p = 0.949$, buying ~72 catalysts plus a control arm for the budget of 17 exhaustive ones.

5

Send the corrected work note

Complete and verified. Every number now traces to a JSON written by a committed script.

6

Run the prospective validation

Prof. Taniike offered to synthesise candidates. A shortlist is a set to test, not a league table — the model cannot order within its own top 20.

Summary

What we asked

Can published OCM literature improve a lab-data model — and can we rank UNSEEN catalysts well enough to guide synthesis?

What we learned

- The original –9.7% gain was catalyst-identity leakage. Under catalyst-grouped CV the same models are 1.8% WORSE than baseline.
- No literature-integration design beats composition alone in-domain — four pre-registered designs and a 28-family follow-up, all null.
- Literature helps only where the lab has NO coverage: on non-impregnation chemistry a plain merge lifts Spearman 0.24 → 0.39 and enrichment 0.42× → 1.34×. Plain merging beats the two-stage machinery — the value is the data, not the method.
- The composition-only model still screens unseen catalysts usefully: enrichment 3.77× on the equal-effort set (95% CI 3.04–4.89×), and that conclusion survives either reading of the open data question.

Next step that matters most

Ask for the reaction-condition columns. One email unlocks more than any modelling change we have tried.